

Basler ace2 a2A4504-18umPRO Monochrome USB3 Pro Camera
#16-635



Basler ace2 USB 3.0 Cameras (Front)

Basler ace2 USB 3.0 Cameras are available in two versions that provide optimized hardware design, proven compact and robust housing, and state of the art CMOS sensor technology with excellent image quality from Sony's 2nd-Gen Pregius, 3rd-Gen STARVIS, or the latest 4th-Gen Pregius S sensors. These second-generation Ace cameras include all of the components of the first-generation Ace cameras along with integrated features such as a status LED, removable IR cut filter, and robust M8 connector. The Basler ace2 USB 3.0 Camera Basic version offers excellent price to performance ratio with powerful Computer Vision feature set, while the Pro version unlocks unique advanced Beyond features such as Pixel Beyond and Compression Beyond for maximum performance to meet with demanding application requirements. These cameras include the industry proven Basler Pylon SDK Software and Driver and conform to GenICam (USB3 Vision) standards, making them a fast and cost-effective solution to software integration. Basler ace2 Cameras are available in both USB 3.0 and GigE interfaces and are ideal for a wide range of applications such as factory automation and industrial robotics.

Spectrum: Monochrome

General

Type:	Monochrome Camera
Model Number:	a2A4504-18umPRO
Manufacturer:	Basler
Camera Series:	ace2

Physical & Mechanical Properties

Dimensions (mm):	36.3 x 29 x 29 (excludes connectors and lens mount)
Weight (g):	<90
Housing:	Full

Sensor

Image Buffer:	Not Specified
Sensor Format:	1.1"
Resolution (Megapixels):	20.20
Frame Rate (fps):	18.00
Pixels (H x V):	4,504 x 4,504
Pixel Size, H x V (µm):	2.74 x 2.74
Sensing Area, H x V (mm):	12.3 x 12.3
Imaging Sensor:	Sony IMX541
Type of Sensor:	Progressive Scan CMOS
Shutter Type:	Global
Pixel Depth:	8/10/12 Bit
Exposure Time:	Freely Programmable
Dynamic Range (dB):	70.8
Machine Vision Standard:	USB3 Vision v1.0, GenICam

Electrical

Power Consumption (W):	4.0
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Hardware & Interface Connectivity

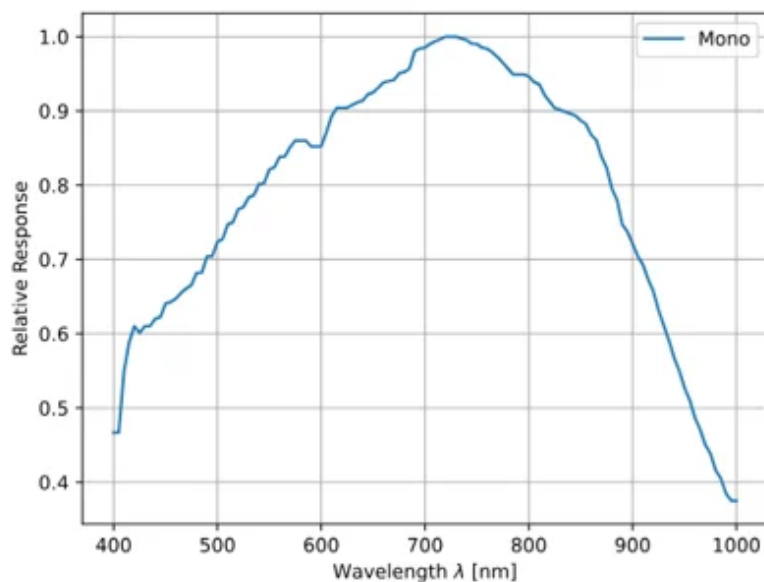
Interface:	USB 3.0
Connector:	USB 3.0, Micro-B
Power Supply:	Power over USB
GPIOs:	1 opto-coupled input line, 2 general purpose I/O (GPIO) lines
Synchronization:	Hardware Trigger (GPIO), Software Trigger, or Free-Run
Interface Port Orientation:	Back Panel
GPIO Connector Type:	6-pin M8

Threading & Mounting

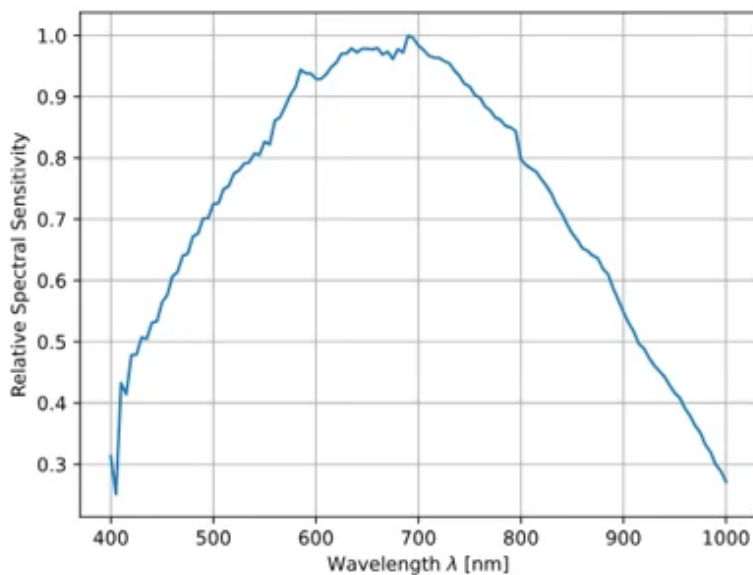
Mount:	C-Mount
Mounting Threads:	1/4-20 with Tripod Mount Adapter #16-641

Environmental & Durability Factors

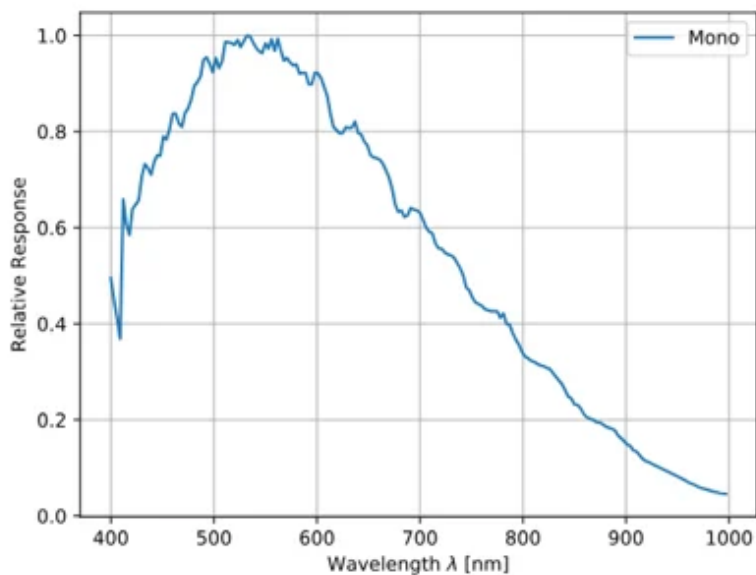
Operating Temperature (°C):	-10 to +60
Storage Temperature (°C):	-20 to +80



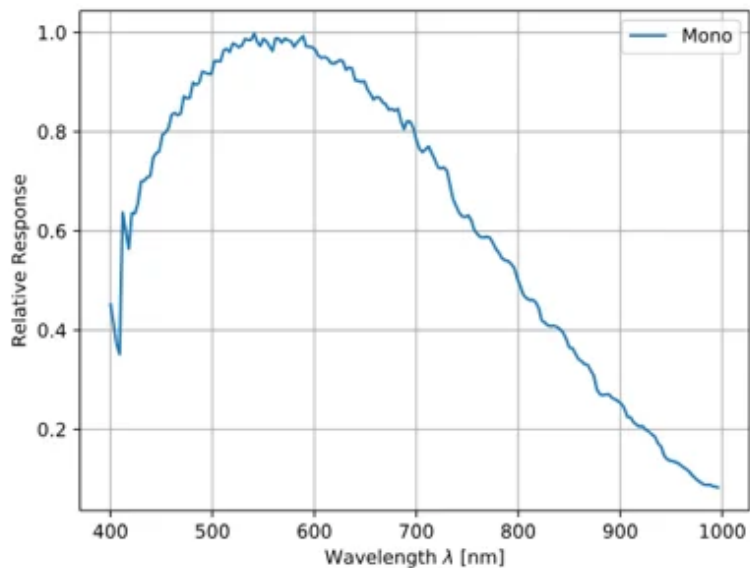
IMX676 Sensor QE Curve



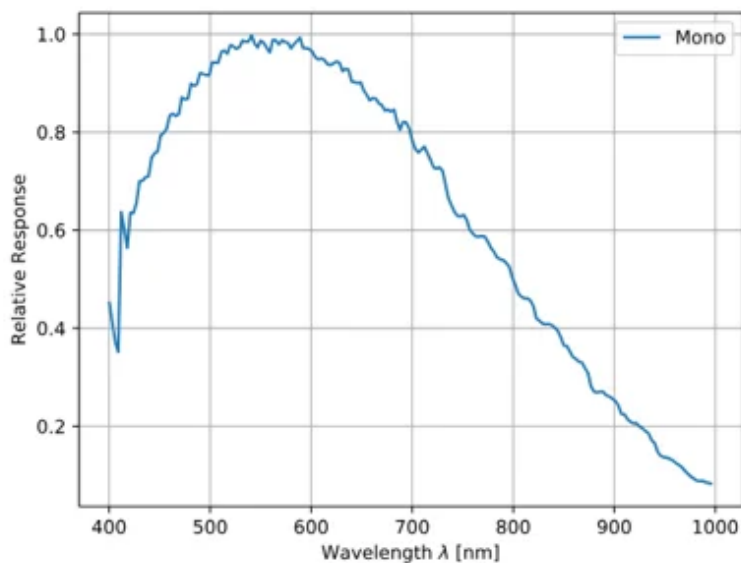
IMX900 Sensor QE Curve



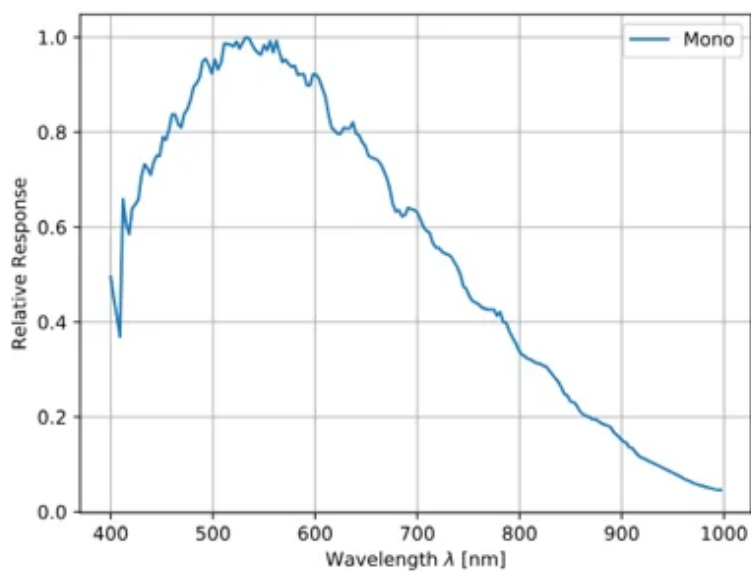
GMAX2505 Sensor QE Curve



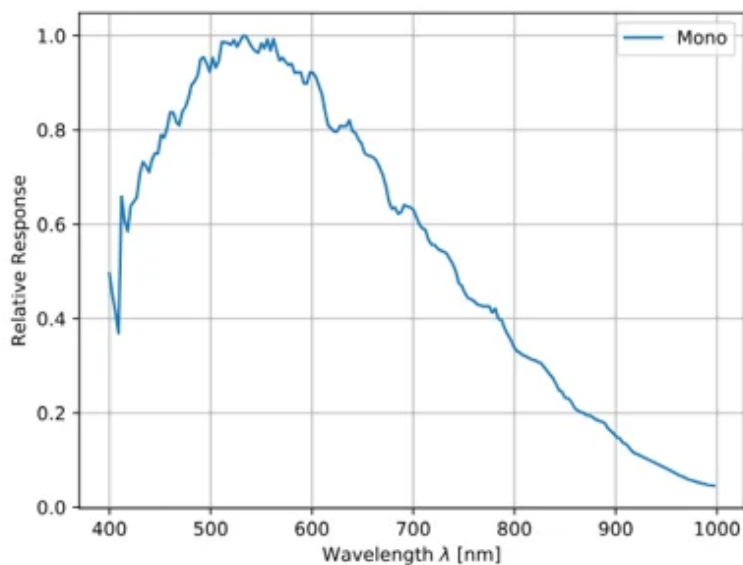
IMX546 Sensor QE Curve



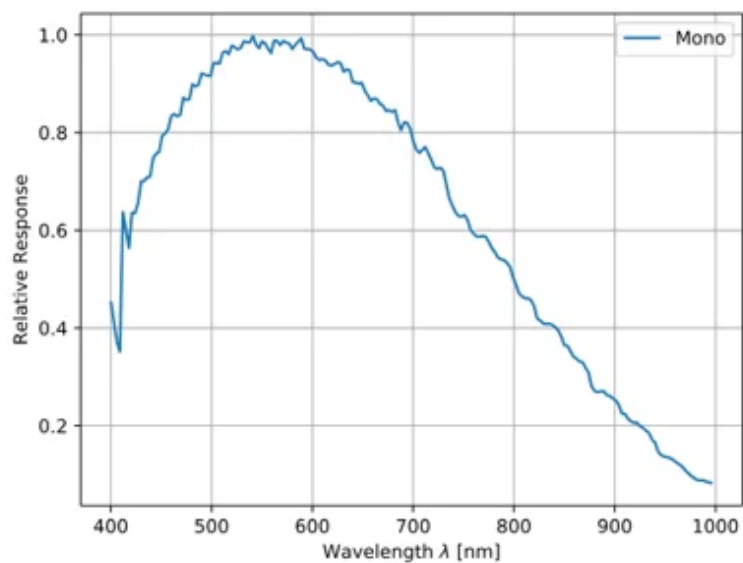
IMX545 Sensor QE Curve



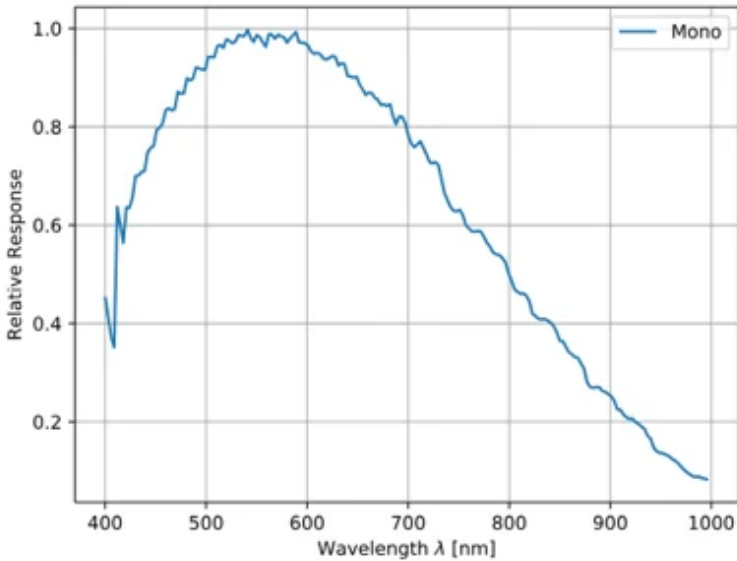
GMAX2509 Sensor QE Curve



GMAX2518 Sensor QE Curve



IMX547 Sensor QE Curve



E2525A Sensor QE Curve



October 07, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

Stock No.	Description
16-635	Basler ace2 a2A4504-18umPRO Monochrome USB3 Pro Camera

This item is EU RoHS (2015/863/EU) compliant without the use of exemptions.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.

Jay Budd, Director of Corporate Compliance
October 07, 2026

Edmund Optics Inc. - 101 E Gloucester Pike, Barrington, NJ 08007 | 1-800-363-1992 | Compliance@edmundoptics.com



October 07, 2026

<i>Stock No.</i>	<i>Description</i>
16-635	Basler ace2 a2A4504-18umPRO Monochrome USB3 Pro Camera

Edmund Optics certifies that all articles included in this shipment are in compliance with the terms and conditions of this order. The company also certifies that the articles included in this shipment are in accordance to all agreed upon specifications and quality assurance provisions. Please call 1-800-363-1992 should any questions arise from this shipment.



Jeff Harvey- EVP of Operations



October 07, 2026

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This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

Stock No.	Description
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This item does not contain any of the substances listed on the Reach SVHC list as of 14 June 2023 in quantities exceeding 0.1% weight by weight.

Please note the following:

- Entry 11 of Annex V of the REACH regulation EC 1907/2006 as amended by regulation 987/2008, exempts “non-hazardous” glass components from registration.
- Items considered to be packing materials and which do not form part of, or are not required for the operation of, the product, are not considered to be within scope of this declaration.
- EO does not routinely analyze products for substances not purposely added or specified within the design specifications, drawings and assembly procedures.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



Jay Budd, Director of Corporate Compliance
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